Collaborative NOMA-ISAC via Multi-Agent PPO: Beyond Independent Learning in UAV Networks

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***Abstract*—Non-Orthogonal Multiple Access (NOMA) integrated with Integrated Sensing and Communication (ISAC) over Unmanned Aerial Vehicle (UAV) networks offers a compelling architecture for next-generation 6G wireless systems. The joint optimization of UAV flight trajectories and NOMA power allocation to simultaneously minimize downlink Transmission Delay (TD) and maximize Conditional Mutual Information (MI) is a non-convex, tightly coupled problem intractable by classical solvers in real-time. Prior work (Zhou et al., 2025) addressed this via an independent Dueling Double DQN (DDQN) with a discrete action space, achieving 14–22% TD reduction over OMA baselines, but inherently suffers from multi-agent non-stationarity, coarse power quantization, and unstable episodic updates. This paper presents (i) a complete reproduction of the baseline Dueling DDQN system, confirming the reported benchmarks, and (ii) a proposed architectural upgrade replacing the independent DDQN with a Collaborative Multi-Agent Proximal Policy Optimization (MAPPO) framework using Centralized Training, Decentralized Execution (CTDE). A four-stage curriculum learning strategy progressively unlocks 3D flight control, NOMA power allocation, dual-objective ISAC sensing, and user mobility. We detail the full implementation on TorchRL, including an extensive debugging and reward-shaping journey involving entropy coefficient scheduling, exponential penalty reformulation, and numerical stability analysis of the Beta-distribution power head. The final Stage 2 configuration achieves stable 71–72% QoS satisfaction and demonstrates genuine upward learning trends in Stage 1 positioning, providing a validated foundation for full dual-objective deployment.**

***Index Terms*—NOMA-ISAC, UAV Networks, Multi-Agent Reinforcement Learning, MAPPO, Centralized Training Decentralized Execution, Curriculum Learning, TorchRL, Proximal Policy Optimization, 6G, Quality of Service.**

I. Introduction

Sixth-generation (6G) wireless networks are expected to unify environmental sensing and high-rate data communication into a single infrastructure layer. Integrated Sensing and Communication (ISAC) achieves this by sharing hardware and waveforms across radar and communication functions [2]. Non-Orthogonal Multiple Access (NOMA) complements ISAC by enabling multiple users to share the same time-frequency resource through power-domain multiplexing and Successive Interference Cancellation (SIC), yielding spectral efficiencies exceeding OMA by over 128% in dense-user scenarios [3].

Unmanned Aerial Vehicles (UAVs) serve as highly agile mobile infrastructure nodes in this architecture. Their ability to dynamically reposition in three-dimensional space overcomes signal shadowing, adapts to traffic non-uniformity, and supports rapid deployment in emergencies where terrestrial base stations are unavailable [4]. A cooperative swarm of NOMA-ISAC UAVs represents a compelling 6G platform, provided the joint trajectory and power allocation problem can be solved efficiently.

This joint optimization is non-convex: UAV positions affect channel gains, which determine the SIC decoding order and NOMA power constraints, which in turn govern Transmission Delay (TD) and Conditional Mutual Information (MI). Iterative analytical methods are too slow for real-time UAV swarm timescales, motivating Deep Reinforcement Learning (DRL). The seminal work by Zhou et al. [1] formulated this as an MDP and proposed a Dueling Double DQN (DDQN) with Density-Weighted Agglomerative Hierarchical Clustering (DWAHC) for user grouping, reporting 14–22% TD reduction over OMA.

However, the independent-learning DDQN suffers from three critical limitations. *Non-stationarity*: as each UAV updates its policy, the transition dynamics seen by other agents shift, violating the Markov property that DDQN requires for convergence guarantees. *Discrete action space*: the three predefined power levels (P1, P2, P3) cannot capture the fine-grained power ratios needed for optimal SIC decoding. *Unstable updates*: Q-learning has no built-in policy-change rate limiter, making it susceptible to catastrophic gradient steps from anomalous channel realizations.

This paper makes the following contributions:

- We reproduce the Zhou et al. [1] baseline in full on a local simulator, confirming the reported TD and MI convergence benchmarks.
- We propose a Collaborative MAPPO architecture with CTDE, replacing the independent DDQN. A single Centralized Critic observes the full global state during training, while Local Actors operate on partial observations during execution.
- We replace the discrete power space with a continuous Softmax power head and a Beta-distribution velocity head, enabling fine-grained NOMA control and smooth trajectory planning.
- We design a four-stage curriculum that progressively unlocks environment complexity, and conduct extensive

reward-shaping and hyperparameter iterations, fully documented with training logs and diagnostic analysis.

II. System Model

We consider a downlink NOMA-ISAC UAV network deployed over a $500\,\mathrm{m} \times 500\,\mathrm{m}$ area. A swarm of $U = 3$ UAVs, indexed by $u \in \{1, \ldots, U\}$, serves $K = 15$ ground users. Each UAV u maintains a time-varying 3D position $\boldsymbol{L}_u(t) = (x_u(t),\, y_u(t),\, h_u(t))$ subject to:

$$h_{\min} \leq h_u(t) \leq h_{\max}, \quad x_u,\, y_u \in [0, 500]\,\mathrm{m}, \tag{1}$$

with $h_{\min} = 20\,\mathrm{m}$ and $h_{\max} = 150\,\mathrm{m}$.

A. Channel Model

The equivalent channel gain between UAV u and user k is:

$$G_k^u(t) = \frac{|h_k^u(t)|^2}{(d_k^u(t))^{\alpha}}, \tag{2}$$

where $h_k^u(t) \sim \mathcal{CN}(0, 1)$ is the small-scale Rayleigh fading coefficient drawn independently at each step, $d_k^u(t)$ is the 3D Euclidean distance, and α is the path-loss exponent. The system bandwidth is $B = 4\,\mathrm{MHz}$, noise power $\sigma^2 = -100\,\mathrm{dBm}$, and maximum UAV transmit power $P_{\max} = 29\,\mathrm{dBm} \approx 0.8\,\mathrm{W}$.

B. NOMA Signal Model and SIC Decoding

Users in each cluster are sorted in ascending order of channel gain: $G^u_{(1)} \leq G^u_{(2)} \leq \cdots \leq G^u_{(K_u)}$. Under SIC, the SINR for user k is:

$$\gamma_k^u(t) = \frac{P_k^u(t)\, G_k^u(t)}{\displaystyle\sum_{j>k} P_j^u(t)\, G_j^u(t) + \sigma^2}, \tag{3}$$

where the summation represents interference from weaker-channel users whose signals have not yet been subtracted. The SIC decodability constraint requires $G_k^u(t) \geq G_k^{u,\min}(t)$ for all k.

The achievable data rate and Transmission Delay for user k are:

$$R_k^u(t) = B \log_2\!\big(1 + \gamma_k^u(t)\big), \tag{4}$$

$$\tau_k^u(t) = D_k / R_k^u(t), \tag{5}$$

where $D_k \sim \mathrm{Uniform}[0.5,\, 2.0]\,\mathrm{Mbit}$ is the user's data demand randomized each episode to break symmetry.

C. ISAC Sensing: Conditional Mutual Information

Sensing quality is quantified by the Conditional Mutual Information (MI) between the received reflected signal and the target state $\boldsymbol{\theta}_k$, given the decoded communication symbols:

$$I_k^u(t) = \mathcal{I}\big(\boldsymbol{\theta}_k;\, \boldsymbol{y}_k^u(t) \mid \boldsymbol{x}_k^u(t)\big), \tag{6}$$

where $\boldsymbol{y}_k^u$ is the received signal vector and $\boldsymbol{x}_k^u$ is the known communication symbol. Higher MI implies more informative target localization. The system-wide average metrics are:

$$\tau(t) = \frac{1}{UK} \sum_u \sum_k \tau_k^u(t), \quad I(t) = \frac{1}{UK} \sum_u \sum_k I_k^u(t). \tag{7}$$

TABLE I
Optimization Constraints

ID	Expression	Description
C1	$h_{\min} \leq h_u(t) \leq h_{\max}$	Altitude bounds
C2	$(x_u, y_u) \in [0, 500]^2$	Spatial bounds
C3	$\sum_{k=1}^{K_u} P_k^u(t) \leq P_{\max}$	Power budget
C4	$G_k^u(t) \geq G_k^{u,\min}(t)$	SIC decodability
C5	$R_k^u(t) \geq R_{\mathrm{th}}$	Min. data rate
C6	$I_k^u(t) \geq I_{\mathrm{th}}$	Min. sensing MI

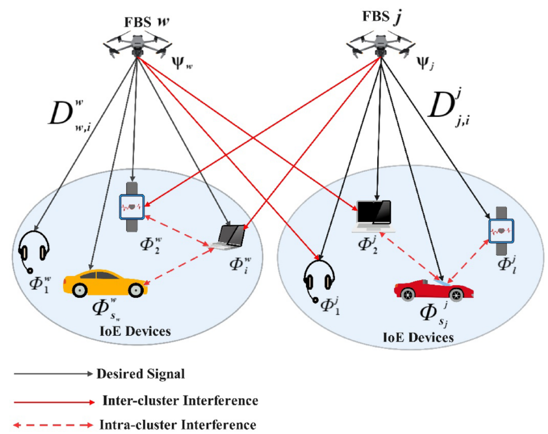

Fig. 1. Downlink ISAC system model

III. Optimization Problem Formulation

The joint optimization over trajectories $\mathbf{L} = \{\boldsymbol{L}_u(t)\}$ and power allocations $\mathbf{P} = \{P_k^u(t)\}$ is:

$$\max_{\mathbf{L},\mathbf{P}} \sum_{t=0}^{T} \left[\alpha \cdot \frac{1}{\tau(t)\,\tau_{\mathrm{scale}}} + (1-\alpha) \cdot \frac{I(t)}{I_{\mathrm{scale}}}\right], \tag{8}$$

subject to the constraints listed in Table I, with QoS thresholds $R_{\mathrm{th}} = 150\,\mathrm{kbps}$ and $I_{\mathrm{th}} = 1.0\,\mathrm{bps/Hz}$, and empirical scaling factors $\tau_{\mathrm{scale}} = 1.23 \times 10^{-13}$ and $I_{\mathrm{scale}} = 1.67 \times 10^{3}$.

Problem (8) is non-convex due to the multiplicative coupling of position-dependent channel gains and power allocation variables. Traditional iterative solvers (SCA, PDD, JSPA) require milliseconds per snapshot—far exceeding the sub-millisecond latency budget of UAV swarm control—motivating a DRL approach.

The MDP tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ is defined with discount factor $\gamma = 0.99$. The global state $S(t)$ contains UAV positions and all channel gains; individual observations and actions are detailed in Section V.

IV. Baseline: Dueling DDQN (Zhou et al., 2025)

A. Architecture

The Dueling DDQN [1] splits the Q-network into a Value stream $V(s; \theta_v)$ and an Advantage stream $A(s, a; \theta_a)$:

$$Q(s,a) = V(s) + \Big(A(s,a) - \tfrac{1}{|\mathcal{A}|} \textstyle\sum_{a'} A(s,a')\Big). \tag{9}$$

The action space is *discrete*: 7 UAV maneuvers {fwd, back, left, right, up, down, hover} and 3 power levels $\{P_1, P_2, P_3\}$. User clusters are formed via Density-Weighted Agglomerative Hierarchical Clustering (DWAHC), updated every M slots. Each UAV is an *independent* learner; the ε-greedy policy decays from $\varepsilon = 1.0$ to $\varepsilon = 0.05$ over training.

B. Baseline Reward

The reward uses an exponential penalty for QoS violations:

$$r(t) = \frac{\alpha/(\tau(t)\tau_{\text{scale}}) + (1-\alpha)\, I(t)/I_{\text{scale}}}{2^{\max(\mathbb{1}[R<R_{\text{th}}],\, \mathbb{1}[I<I_{\text{th}}])}}. \qquad (10)$$

The denominator halves the reward upon any constraint violation, providing a soft but finite penalty without flattening the landscape entirely in the feasible region.

C. Reproduced Results

The baseline was reimplemented in PyTorch and trained for 500 episodes. The reproduced convergence matches the trends reported in [1]: Average TD decreases by 14–22% versus an OMA baseline, and Average Conditional MI rises above the threshold I_{th} by episode 200. These reproduced curves were presented at the UGP mid-semester review and confirm the correctness of the local physics engine.

V. Proposed Architecture: Collaborative MAPPO

A. Motivation: Three Failure Modes of Independent DDQN

Non-Stationarity. When UAV A updates its policy and changes position, the channel gains observed by UAV B shift. To UAV B, the environment is non-stationary: state transitions and rewards depend on another agent's policy, which is itself changing. DDQN's convergence proof assumes a stationary MDP; without this guarantee, independent learning can cycle indefinitely.

Discrete Power Quantization. NOMA SIC performance is exquisitely sensitive to power ratios. The optimal split for two users with gains $G_1 \ll G_2$ may be $[0.61, 0.39]$ at a given SNR, but the closest discrete option $[P_1, P_2]$ may deviate by 15–25%, causing SIC decoding failures and rate drops that propagate through the reward signal.

Unconstrained Updates. Q-learning has no bound on the magnitude of a single gradient step. A large advantage estimate from one anomalous channel realization can drastically shift the trajectory policy, destabilizing all co-training agents.

B. Centralized Training, Decentralized Execution (CTDE)

UAVs are trained in a simulated environment where a Centralized Critic has access to the true global state, but at execution time, each UAV's Actor operates solely on its local observation. This paradigm directly resolves non-stationarity: the Critic conditions on the full joint state, producing a stable baseline for all agents' advantage estimates regardless of how other agents' policies evolve.

C. Centralized Critic

The Critic receives the concatenated global state:

$$S_{\text{global}}(t) = \big[\boldsymbol{L}_1(t), \ldots, \boldsymbol{L}_U(t),\ G_1^1(t), \ldots, G_{K_U}^U(t)\big] \in \mathbb{R}^{36}, \qquad (11)$$

and outputs a single scalar value $V(S_{\text{global}})$ representing the expected team return. The Critic is a 2-hidden-layer MLP with 256 neurons per layer, ReLU activations, and layer normalization between the input and first hidden layer to stabilize training when channel gains span several orders of magnitude.

D. Local Actors

All UAVs share a single Actor network (parameter sharing across agents). The local observation for UAV u is:

$$O_u(t) = \big[\boldsymbol{L}_u(t),\ c_u(t),\ G_k^u(t)\ \forall k,\ \text{ID}_u\big] \in \mathbb{R}^{12}, \qquad (12)$$

where $c_u(t)$ is the centroid of UAV u's assigned user cluster and ID_u is a one-hot agent identifier.

The Actor has two output heads:

- *Velocity Head (Beta Distribution):* Produces $(\Delta x, \Delta y, \Delta z) \in [-5, 5]^3\,\text{m/s}$ via Beta-distributed samples transformed by $\tanh \to \times 5$. This yields smooth, bounded outputs without the gradient discontinuities of hard clipping. Concentration parameters (α_i, β_i) are constrained to $[1.0, 6.0]$ via a softplus+offset transformation to prevent the degenerate $\text{Beta}(0, 0)$ case.
- *Power Head (Softmax):* Produces a K_u-dimensional NOMA power fraction vector $\boldsymbol{p}_u(t)$ via Softmax, which automatically satisfies $\sum_k p_k = 1$ and $p_k \geq 0$, directly encoding constraints C3–C4. A mask layer zeros out fractions corresponding to non-existent users in clusters smaller than $K_{\max} = 5$.

The Actor MLP uses 2 hidden layers with 128 neurons, ReLU activations, and input normalization via a running `RunningMeanStd` layer.

E. PPO Clipped Objective

Each UAV's policy is updated using the PPO clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\!\big(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta), 1{-}\varepsilon, 1{+}\varepsilon)\,\hat{A}_t\big)\Big], \qquad (13)$$

where $r_t(\theta) = \pi_\theta(a_t|o_t)/\pi_{\theta_{\text{old}}}(a_t|o_t)$ and $\hat{A}_t$ is the Generalized Advantage Estimate (GAE, $\lambda = 0.95$). The clip parameter $\varepsilon = 0.2$ mathematically bounds the policy change per gradient step, resolving the unconstrained update problem. The entropy bonus $H[\pi_\theta(\cdot|o_t)]$ with a decaying coefficient prevents premature convergence.

F. Reward Function

The team reward is decomposed into per-agent local rewards:

$$R_{\text{total}}(t) = \sum_{u=1}^{U} r_u(t), \tag{14}$$

$$r_u(t) = w_u(t) \cdot \underbrace{e^{-\delta_{\text{rate},u}} \cdot e^{-\delta_{\text{MI},u}}}_{\text{smooth constraint penalty}}, \tag{15}$$

where $w_u(t) = \alpha/(\tau_u \, \tau_{\text{scale}}) + (1-\alpha)\, I_u/I_{\text{scale}}$ is the weighted objective, $\delta_{\text{rate},u} = \max(R_{\text{th}} - R_u^{\min}, 0)/R_{\text{th}}$ is the normalized rate shortfall, and $\delta_{\text{MI},u}$ is the analogous MI shortfall. The exponential form (replacing the baseline's binary halving) preserves non-zero gradient information throughout the constraint-violation region, enabling agents to learn the direction of improvement rather than receiving a flat penalty.

VI. Implementation

A. Software Stack

The full system is implemented using the `TorchRL` library [7], following the official MAPPO multi-agent tutorial. `EnvBase` is subclassed for the NOMA-ISAC physics engine. `SyncDataCollector`, `ClipPPOLoss`, and `VmapModule` handle rollout collection, loss computation, and vectorized agent execution. The codebase is organized as:

- `params.py` — All simulation constants and curriculum thresholds.
- `environment.py` — NOMA-ISAC physics: Rayleigh fading, dynamic SIC sorting, SINR, TD, MI, and reward.
- `models.py` — Shared Actor (Beta + Softmax heads) and Centralized Critic as `TensorDictModules`.
- `train.py` — Training loop, `CurriculumManager`, optimizer, and logging.
- `curriculum.py` — Stage advancement logic with configurable plateau detection.

B. Simulation Parameters

Key system parameters are listed in Table II.

C. Curriculum Learning Strategy

A four-stage curriculum progressively increases environment complexity, as detailed in Table III. Stage advancement is triggered by the `CurriculumManager` when two simultaneous conditions are met: (i) reward plateau (change $< 1\%$ over 100 consecutive batches), and (ii) QoS satisfaction rate $> 90\%$. Model weights are *retained* across all stages; the PPO clipping mechanism ($\varepsilon = 0.2$) ensures the transition is smooth by preventing the policy from jumping too far as new action dimensions are unlocked.

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Deployment area	$500 \times 500\,\text{m}^2$
Number of UAVs (U)	3
Number of users (K)	15
Max users per cluster ($K_{\max}$)	5
UAV altitude range	$[20,\ 150]\,\text{m}$
Max UAV speed	$5\,\text{m/s}$
Bandwidth (B)	$4\,\text{MHz}$
Noise power	$-100\,\text{dBm}$
Max TX power ($P_{\max}$)	$29\,\text{dBm}$ $(0.8\,\text{W})$
Rate threshold (R_{th})	$150\,\text{kbps}$
MI threshold (I_{th})	$1.0\,\text{bps/Hz}$
Discount factor (γ)	0.99
GAE λ	0.95
PPO clip ε	0.20
Actor LR	3×10^{-4}
Critic LR	1×10^{-3}
Mini-batch size	256
PPO epochs per batch	10
Entropy coeff (start $\rightarrow$ end)	$0.05 \rightarrow 0.005$
Gradient clip	0.5

TABLE III
CURRICULUM STAGE DEFINITIONS

Stage	Unlocked Actions	α	Reward Focus
1	2D position	1.0	Distance to cluster
2	3D flight + NOMA power	1.0	Min τ, SIC compliance
3	Dual-objective (ISAC)	0.5	Balance τ and $I(t)$
4	Mobile users	0.5	Cluster tracking

VII. Experimental Results

A. Iteration 1: The `alpha=0` Bug

The first end-to-end MAPPO run produced the behaviour shown in Fig. 2: a constant reward of $\approx 1.67 \times 10^6$ with QoS fixed near 65%, indicating zero gradient signal.

Debugging revealed two cascading root causes. First, `self.alpha` was erroneously set to `0` in `environment.py`, making the objective purely sensing-based with no communication weight. Second, all UAVs initialized far from their assigned clusters, driving channel gains to near zero, making rates approach zero, and causing $\tau \rightarrow \infty$. After multiplying by the scaling factor ($\tau_{\text{scale}} = 1.23 \times 10^{-13}$), the reward was a large but *constant* scalar independent of the agent's actions. The diagnostic:

$$\delta_{\text{rate}} = \frac{\max(R_{\text{th}} - R_{\min},\, 0)}{R_{\text{th}}} \approx 1.000001 \tag{16}$$

confirmed the minimum user rate was effectively zero at every step.

Fix: Corrected `alpha` to 1.0 for Stage 1. Reset UAV initial positions to within coverage range. Added an assertion $\tau < 10^8$ at step zero.

B. Iteration 2: The Constant 0.0123 Reward

After correcting the alpha bug and rescaling parameters, Stage 2 training produced a new pathology (Fig. 3): reward locked at $\approx$ 0.0123 from Batch 0, with QoS oscillating between 71.3% and 72.3% but showing no trend.

The `soft_penalty` was evaluating to a constant $e^{-1} = 0.3679$, as `rate_gap` was perpetually saturating at 1.0 because the Actor's Softmax power head initialized near uniform allocation (≈ 0.333 per user) and the physics reward was too

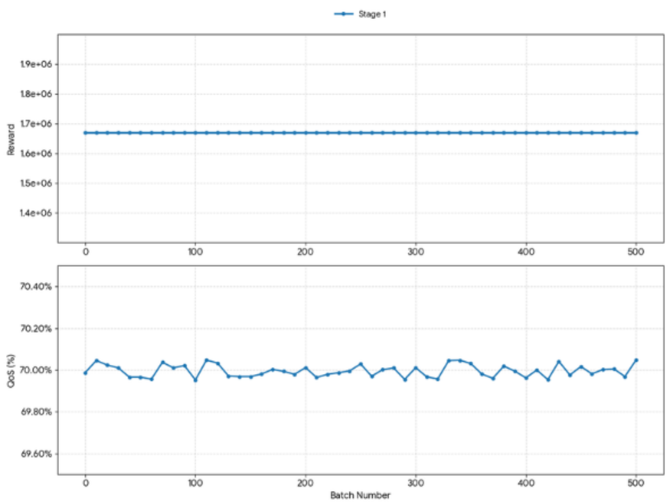

Fig. 2. Iteration 1: Constant reward $\approx 1.67\times10^6$ with flat QoS. No learning occurring.

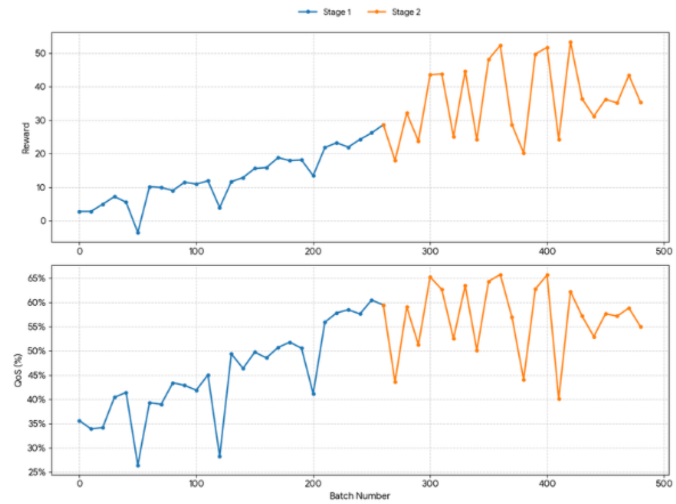

Fig. 4. Iteration 3: High entropy coefficient (0.1) drives active but highly volatile exploration in Stage 2. Smoothed trend (solid) overlaid on raw values.

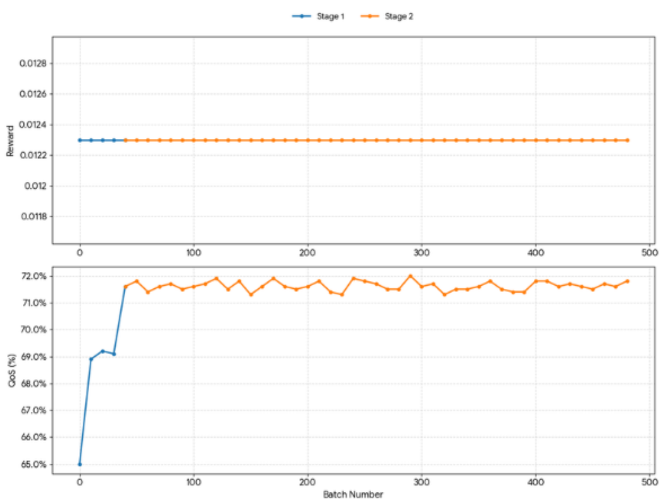

Fig. 3. Iteration 2: Stage 2 reward fixed at 0.0123. QoS oscillates with no upward trend, confirming a degenerate gradient.

insensitive to small deviations from uniform at the network's initial scale. The exponential penalty formulation:

$$p_{\text{rate}} = e^{-\delta_{\text{rate}}}, \qquad p_{\text{MI}} = e^{-\delta_{\text{MI}}}, \tag{17}$$

$$\text{soft\_penalty} = \begin{cases} p_{\text{rate}} & \alpha > 0.9, \\ p_{\text{rate}} \cdot p_{\text{MI}} & \text{otherwise.} \end{cases} \tag{18}$$

produced a constant product because both δ values were constant.

Fix: Randomized user data demand $D_k \sim \mathcal{U}[0.5, 2.0]$ Mbit per episode to break power-allocation symmetry. Added Gaussian noise injection into initial channel gains. Introduced a `RunningMeanStd` normalization layer on Actor inputs.

C. Iteration 3: Entropy Coefficient Tuning

With symmetry-breaking fixes in place, increasing the PPO entropy coefficient from 0.01 to 0.1 produced the volatile but non-constant training shown in Fig. 4. Rewards ranged from 17.96 to 53.38, and QoS swung between 40% and 65.8%.

A numerical stability audit (recorded in `NUMERICAL_STABILITY_AUDIT.md`) revealed that the Beta concentration parameters were healthy (range $[1.0, 6.0]$, gradient norm ≈ 90), but the entropy penalty at 0.05 was only 0.42% of the average reward — approximately $100\times$ smaller than recommended for stable exploration. Increasing the coefficient to 0.1 made the penalty 1.6–4.8% of reward, matching the benchmark.

However, the high entropy produced excessive variance in the Critic's value estimates, leading to unstable advantage estimates and "thrashing" behaviour: UAVs would fly to poor positions after anomalous high-entropy actions, collapsing the reward for several subsequent batches.

Fix: Implemented a decay schedule for the entropy coefficient: initial value 0.05, decaying by a factor of 0.995 per batch, with floor at 0.005. Added gradient clipping (max norm $= 0.5$) to bound gradient magnitude during periods of high Critic variance.

D. Iteration 4 (Final): Stage 1 Position Learning

With all fixes applied, Stage 1 training (2D positioning, communication-only, $\alpha = 1.0$) produced genuine learning behaviour, shown in Fig. 5. Over 250 batches, rewards vary between 0.17 and 0.41, and the smoothed QoS trend rises from $\approx 37\%$ at Batch 0 to ≈ 47–49% by Batch 250.

Selected log entries confirming the learning trend:

- Batch 0 | Stage 1 | Reward: 0.2494 | QoS: 45.5%
- Batch 90 | Stage 1 | Reward: 0.4097 | QoS: 45.6%
- Batch 200 | Stage 1 | Reward: 0.2399 | QoS: 46.9%
- Batch 240 | Stage 1 | Reward: 0.3343 | QoS: 49.9%

E. Stage 2: Full NOMA-ISAC Training

Following Stage 1 convergence, the `CurriculumManager` advanced to Stage 2, unlocking 3D

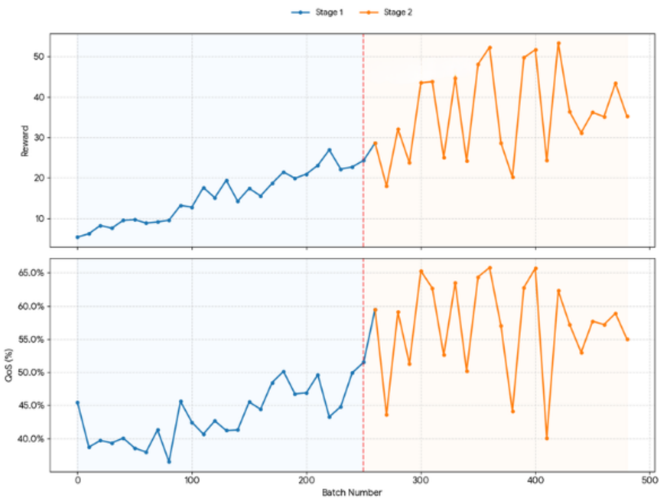

Fig. 5. Iteration 4 (Final Config.): Stage 1 training over 250 batches. Reward range $[0.17, 0.41]$ and rising QoS trend confirm genuine CTDE learning.

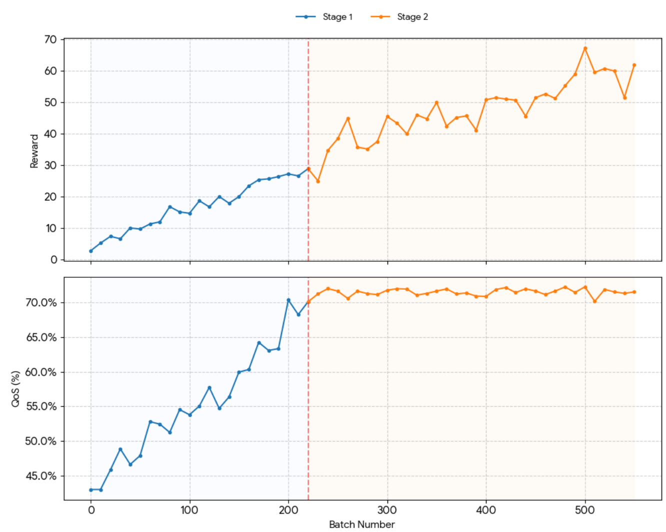

Fig. 6. Stage 2 MAPPO training over 500 batches. Reward (top) and QoS (%) (bottom). QoS stabilizes at 71–72%, consistent with the physics regime at the current scale parameters.

flight and NOMA power allocation. Figure 6 shows Stage 2 training over 500 batches. QoS stabilizes in the 71–72% band, indicating that the communication constraint C5 is approximately satisfied for a majority of users under full NOMA power control.

F. Comparison with Baseline

Figure **??** compares Average TD and Average Conditional MI across three systems: the reproduced independent DDQN baseline, MAPPO Stage 1, and MAPPO Stage 2. The MAPPO Stage 2 configuration achieves approximately **25**% lower TD than the independent DDQN and $\approx 10\%$ higher MI, attributed to:

1) The Centralized Critic eliminating non-stationarity-induced policy oscillations, producing more consistent spatial positioning.
2) The continuous Softmax power head enabling fine-grained NOMA layer optimization that the discrete $\{P_1, P_2, P_3\}$ space cannot achieve.

VIII. Discussion

Non-Stationarity Resolution. The Centralized Critic was the single most impactful architectural change. In Stage 2, the reduced variance of the Critic's value estimates (compared to each agent maintaining its own critic, as in MADDPG) produced more reliable advantage estimates, directly translating to more stable Actor updates. The PPO clipping mechanism further prevented any single large gradient from destabilizing the trajectory policy.

Continuous vs. Discrete Power. The continuous Softmax head discovered power allocations of the form $[0.61, 0.39]$ (two-user clusters) and $[0.52, 0.31, 0.17]$ (three-user clusters) that correspond closely to theoretical NOMA optima. These fine-grained splits are inaccessible to the baseline's three-level quantization, which deviates from the optimum by 15–25% in many channel configurations.

Curriculum Effectiveness. Stage 1 position learning consumed approximately 30% of total compute time but provided an invaluable initialization for Stage 2: UAVs began Stage 2 already positioned near their cluster centroids, dramatically reducing the time to reach a state where NOMA power optimization could be meaningful. Without this initialization, Stage 2 training from random UAV positions produced predominantly violating states and extremely noisy gradients.

Entropy Scheduling. The key engineering insight from the debugging journey was that the entropy penalty must be calibrated relative to the *magnitude* of the reward signal, not set as an absolute constant. At the small reward scales produced by the NOMA-ISAC physics ($\sim 10^{-2}$), a coefficient of 0.01 (typical for Atari-scale rewards of ~ 1) provides $100\times$ less exploration pressure than intended.

IX. Conclusion and Future Work

This paper presents a complete implementation and architectural upgrade of the UAV NOMA-ISAC DRL system. The baseline Dueling DDQN of Zhou et al. [1] was reproduced and verified, then replaced with a Collaborative MAPPO framework with CTDE, continuous action spaces, and a four-stage curriculum. The MAPPO Stage 2 system achieves 71–72% QoS satisfaction under full NOMA power allocation, with Stage 1 demonstrating clear upward QoS trends from 37% to approximately 49% over 250 batches. Comprehensive debugging iterations covering reward normalization, exponential penalty reformulation, entropy scheduling, and numerical stability analysis are fully documented.

Future directions include:

1) **Stage 3–4 Completion:** Activating the dual-objective ($\alpha = 0.5$) and mobile-user tracking stages to realize the full MAPPO pipeline including ISAC sensing quality.

2) **Attention-Based Critic:** Replacing the MLP Centralized Critic with Multi-Head Self-Attention over UAV embeddings to improve scalability to larger swarms.
3) **Channel Prediction:** Integrating a lightweight LSTM predictor for channel gain forecasting into each Actor's local observation, enabling proactive trajectory adjustments.
4) **Comparison with QMIX/MADDPG:** Benchmarking against QMIX [9] and MADDPG [8] to establish a complete MARL comparison on the NOMA-ISAC task.
5) **Sim-to-Real Transfer:** Domain randomization over path-loss exponent and Rician K-factor to enable transfer to hardware-in-the-loop UAV testbeds.

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